

Zhuoqian (Zack) Yang

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EDUCATION

École Polytechnique Fédérale de Lausanne, IVRL

Lausanne, CH

Sep 2023 – Present

- PhD Candidate, supervised by *Dr. Mathieu Salzmann* and *Prof. Sabine Süsstrunk*
- Thesis Title: *Weight Space Representation Learning*

Carnegie Mellon University, Robotics Institute

Pittsburgh, PA, US

Aug 2019 – Dec 2020

- M.S. in Computer Vision, advisor: *Prof. Kris Kitani* | GPA: 4.03/4.3
- Courses: Computer Vision (A), Computer Graphics (A), Machine Learning (A+), Geometry-Based Computer Vision (A+), Computational Photography (A-), Learning-Based Computer Vision (A-), Math for Robotics (A)

Beihang University, School of Software Engineering

Beijing, CN

Sep 2015 – Jun 2019

- B.E. in Software Engineering | GPA: 88.1/100, Ranking: 6/149

PUBLICATIONS

Full list available at <https://scholar.google.com/citations?user=WTDthdcAAAAJ>.

* equal contribution

[1] **Weight Space Representation Learning via Neural Field Adaptation,**

CVPR 2026

Zhuoqian Yang, Mathieu Salzmann, Sabine Süsstrunk

Project Page: <https://inrainbws.github.io/wsr/>

[2] **Subtractive Modulative Network with Learnable Periodic Activations,**

ICASSP 2026 (Oral)

Tiou Wang, Zhuoqian Yang, Markus Flierl, Mathieu Salzmann, Sabine Süsstrunk

Project Page: <https://inrainbws.github.io/smn/>

[3] **3DHumanGAN: 3D-Aware Human Image Generation with 3D Pose Mapping,**

ICCV 2023

Zhuoqian Yang, Shikai Li, Wayne Wu, Bo Dai,

Project Page: <https://3dhumangan.github.io/>

[4] **OrthoPlanes: A Novel Representation for Better 3D-Awareness of GANs,**

ICCV 2023

Honglin He*, Zhuoqian Yang*, Shikai Li, Bo Dai, Wayne Wu

Project Page: <https://orthoplanes.github.io/>

[5] **TransMoMo: Invariance-Driven Unsupervised Video Motion Retargeting,**

CVPR 2020

Zhuoqian Yang*, Wentao Zhu*, Wayne Wu*, Chen Qian, Qiang Zhou, Bolei Zhou, Chen Change Loy,

Project Page: <https://inrainbws.github.io/transmomo>

[6] **MEAD: A Large-Scale Audio-Visual Dataset for Emotional Talking Face Generation,**

ECCV 2020

Kaisiyuan Wang*, Qianyi Wu*, Linsen Song*, Zhuoqian Yang, Wayne Wu, Chen Qian, Ran He, Yu Qiao, Chen Change Loy,

Project Page: <https://wywu.github.io/projects/MEAD/MEAD.html>

RESEARCH

Shanghai AI Laboratory, Research Engineer

Shanghai, CN

EXPERIENCE

Topic: 3D-Aware Representation and Generation

Supervisor: *Dr. Bo Dai* and *Dr. Wayne Wu*

Mar 2021 - Jul 2023

- Proposed one of the first 3D-aware generative adversarial network 3DHumanGAN that synthesizes full-body human images; achieved pose-conditioned 3D-aware synthesis of full-body images with image quality comparable to that of 2D GANs.
- Proposed a novel hybrid implicit-explicit volumetric representation Orthoplanes for neural rendering of 3D scenes; significantly improves quality and view-consistency of rendering on large view-angles.

CMU and Fujitsu Research of America, Master's Student

Pittsburgh, PA, US

Topic: Semantic Facial Image Manipulation using 2D/3D Modalities, *sponsored capstone project*
 Supervisor: *Dr. Laszlo Jeni* and *Dr. Koichiro Niinuma* May 2020 - Dec 2020
 Presentation: <https://drive.google.com/file/d/1UbTcLudyGlr4OiP10rQ53OjattXGkA5B>

- Built a facial expression manipulation model to generate photorealistic images conditioned on FACS.
- Designed a two-stage pipeline: (i) manipulate image geometry using 3D information of the face, (ii) synthesize facial-expression-induced textures such as wrinkles.
- Achieved 19% improvement in FACS Expression Accuracy and 12% improvement in FID.

SenseTime, Research Intern Beijing, CN
Topic: Video Motion Transfer
 Supervisor: *Dr. Wayne Wu* May 2019 - Nov 2019

- Designed an autoencoder framework to learn latent representations of human motion from unpaired videos.
- Achieved unsupervised representation disentanglement by exploiting invariance properties of three orthogonal factors of variation including motion, structure, and view-angle.
- Achieved motion retargeting MSE 20% smaller than the supervised SOTA with our unsupervised method.

Intelligent Computing and Machine Learning Lab, Beihang University, Research Student Beijing, CN
Topic: Visual Question Answering
 Supervisor: *Prof. Zengchang Qin* Jul 2018 - Dec 2018

- Designed a graph neural network approach to enable agents to reason visual relationships on scene graphs.
- Introduced prior knowledge of visual relationships via contrastive-learned embeddings constrained by visual context and language priors.
- Improved performance in questions that require relational reasoning.

Engineering Research Center of GIS Technology, Yunnan Normal University, Research Student Kunming, CN
Topic: Non-Rigid Image Registration
 Supervisor: *Prof. Yang Yang* Mar 2017 - Dec 2017

- Designed a dynamic Gaussian component density to progressively exploit available image information and provide sufficient reliable correspondences for image registration.
- Devised a VGG-pyramid-feature-based non-rigid image registration method;
- Devised a space curvature preservation to improve the plausibility of estimated transformation.
- Improved RMSE by 20% on remote sensing images with temporal appearance changes.

AWARDS EDIC PhD Fellowship, CHF 54'000, EPFL Mar 2023
 First Prize Scholarship for Academic Performance (Top 5%), Beihang University Dec 2016

ACADEMIC SERVICES **Reviewer:** CVPR, ECCV, AAAI, ICLR, ACM MM
Volunteer, ICCP 2024 Lausanne, CH

Teaching Assistant, Beihang University
 Course *Introduction to Software Engineering* Mar 2018 – Jun 2018
Teaching Assistant, EPFL
 Course *CS-341 Computer Graphics* Jan 2024 – May 2024
Teaching Assistant, EPFL
 Course *CS-233 Introduction to Machine Learning* Jan 2026 – May 2026

TALKS **Weight Space Representation Learning via Neural Field Adaptation** Feb 2026
 AI:ML Group Seminar, University of St. Gallen